TRY HACK ME

MOBILE ACQUISITION

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COURSE: COMPUTER FORENSICS AND INVESTIGATION

ASSIGNMENT: LAB ASSIGNMENT

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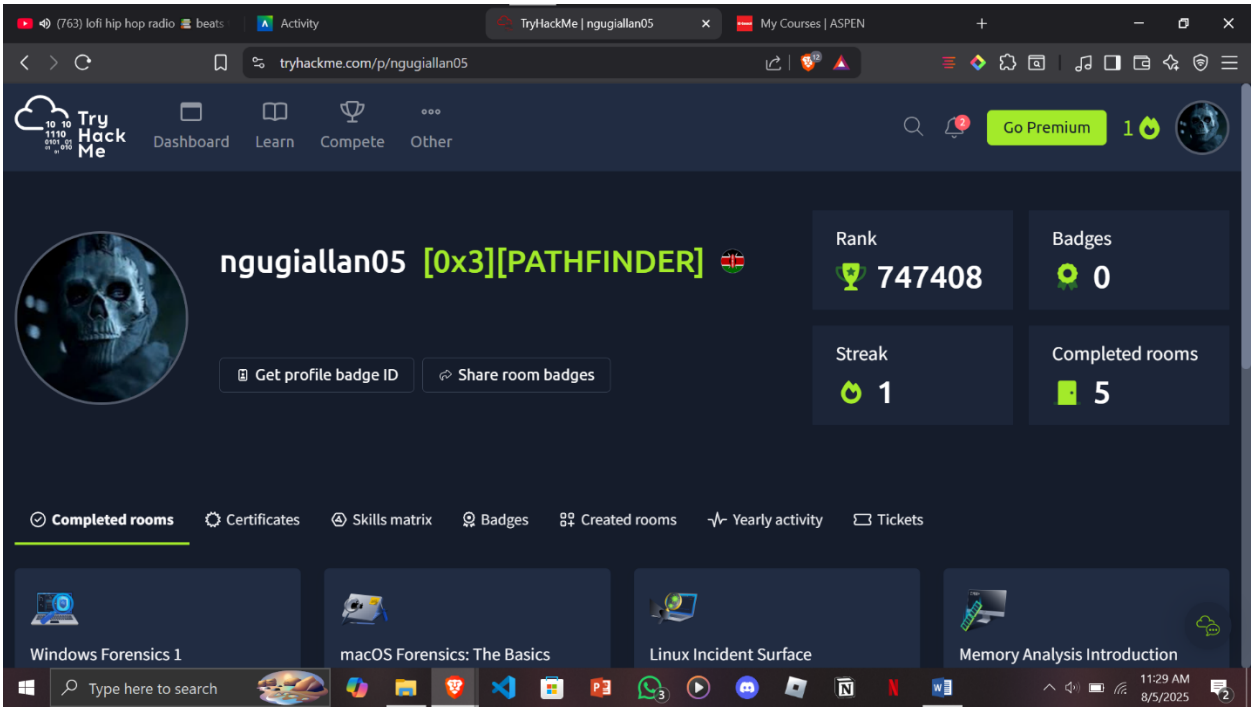
PROFILE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Windows Forensics 1
macOS Forensics: The Basics
Linux Incident Surface
Memory Analysis Introduction
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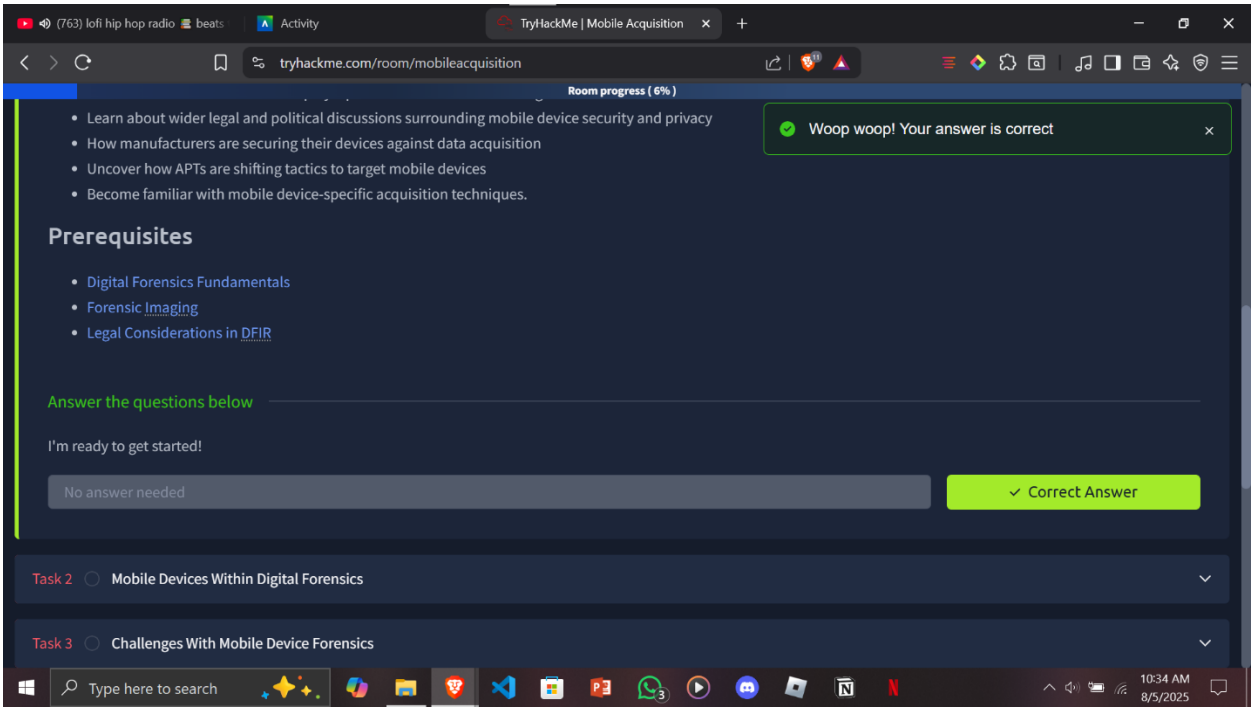
Task 1: Introduction

For digital forensic investigators, mobile devices have turned into a treasure trove. They can provide a wealth of behavioral, communication, and personal information and are utilized on a regular basis.

In this room, one learns how mobile devices can hold important evidence throughout an investigation in this room. Learn about the common issues that make these investigations challenging, how these devices fit into the larger digital forensics process, and why threat groups—like APTs—are focused more on them.

One also looks at a variety of data collection methods, the legal debates occurring in the larger field, and how mobile device manufacturers safeguard their products. Lastly, one performs a capture on a mobile device, joining the ranks of TryAnalyseMe.



Key Takeaway

This first task gives one a briefing about mobile acquisition, presenting smartphones as digital vaults brimming with behavioral and personal information. It reiterates the importance of mobile evidence, the technical and legal challenges of extracting it, and why advanced threat actors are

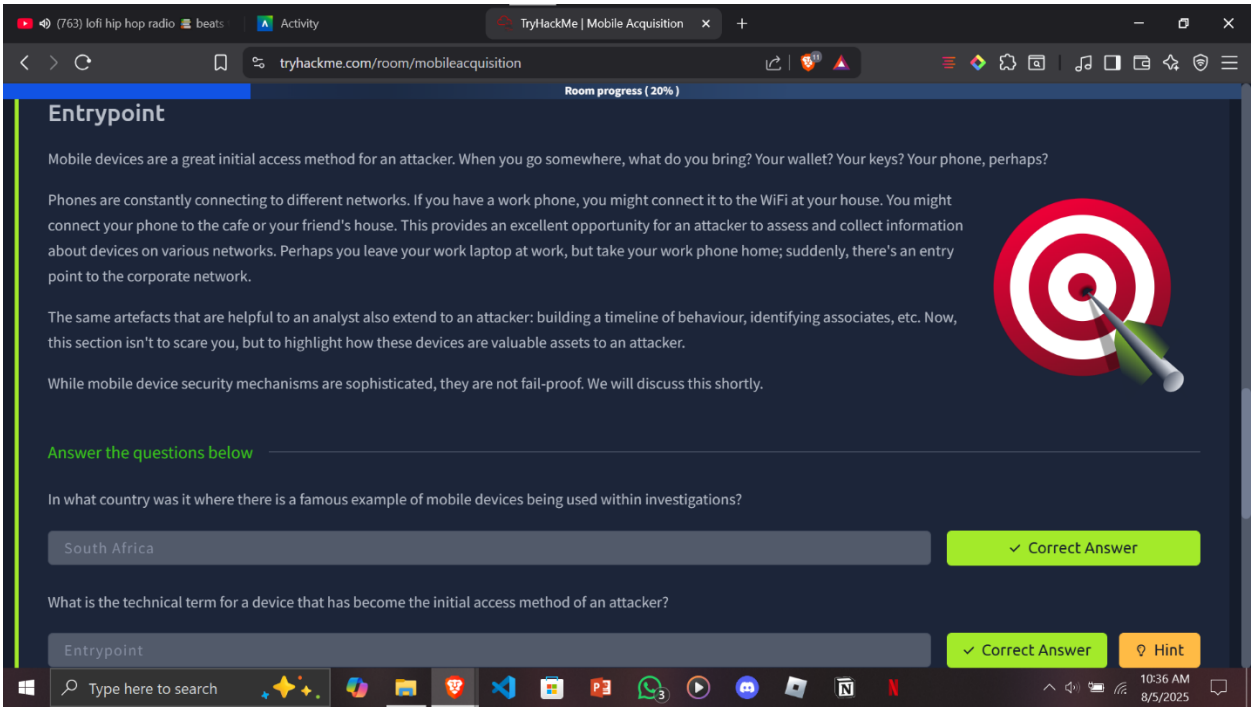
increasingly targeting these devices. It also previews the key subject areas of the room, including data acquisition methods, legal matters, and day-to-day forensic practice.

Task 2: Mobile Devices Within Digital Forensics

In the modern world, mobile smartphones are a strong new technology that is being used all over the world. Some of our most private information and interactions are stored on these mobile devices, like smartphones, which are ingrained in our everyday lives. For an analyst, they are a veritable gold mine of possible evidence.

Since mobile devices can now do computations as efficiently as desktops and laptops, they contain a wealth of priceless artifacts such as call and chat logs, Wi-Fi history amongst others.

After completion of the task, questions were provided, of which they were answered correctly. The questions, together with the answers, are shown in the image below:



Key Takeaway

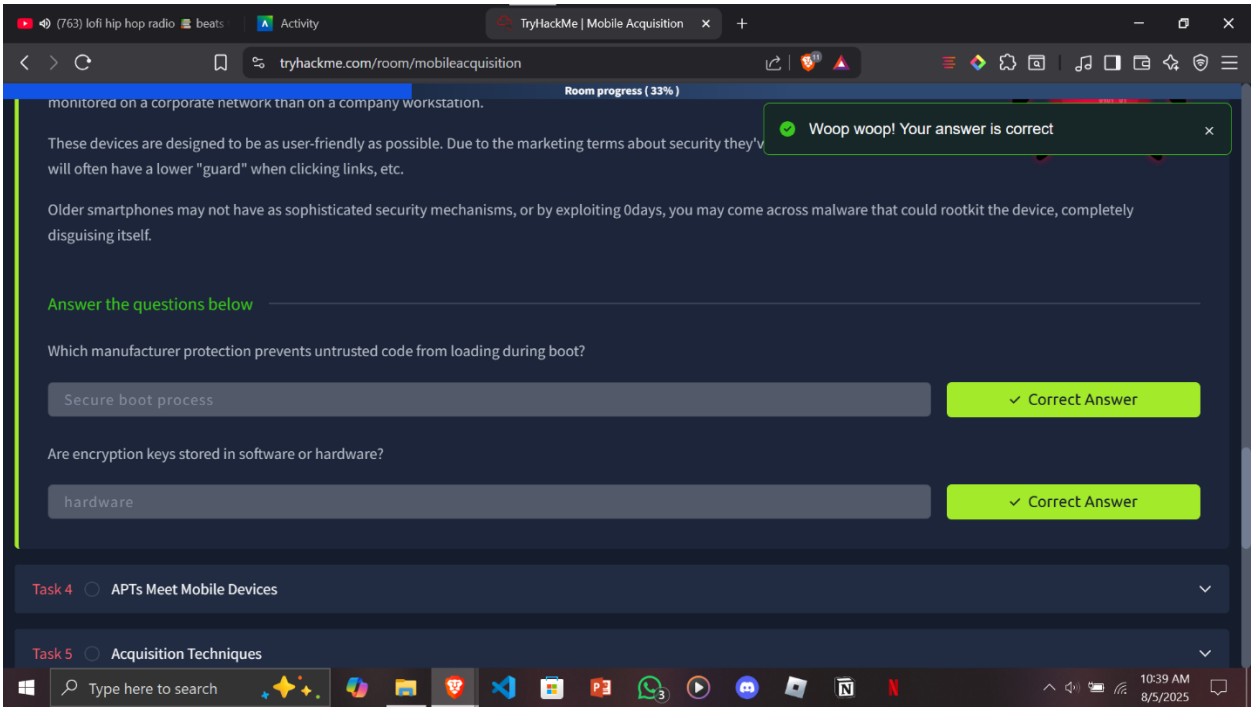
A real-world instance in South Africa is provided, where step-count data from a phone formed a key part of disproving a defendant's alibi, indicating the potential of mobile data as circumstantial evidence. Another key takeaway is that an "entrypoint" is a technical term used to

describe a device used as an initial-contact vector by attackers traversing networks. It indicates the extent to which ubiquitous personal devices can become focal security threats.

Task 3: Challenges with Mobile Device Forensics

This exercise highlights the difficulties faced by forensic examiners, particularly due to security measures on cell phones. It covers challenges such as encryption, secure boot, and limited physical access. These restrictions highlight careful planning and selecting proper tools while performing mobile forensic examination.

Following the completion of the task, there were question answered correctly. This is shown in the image below:



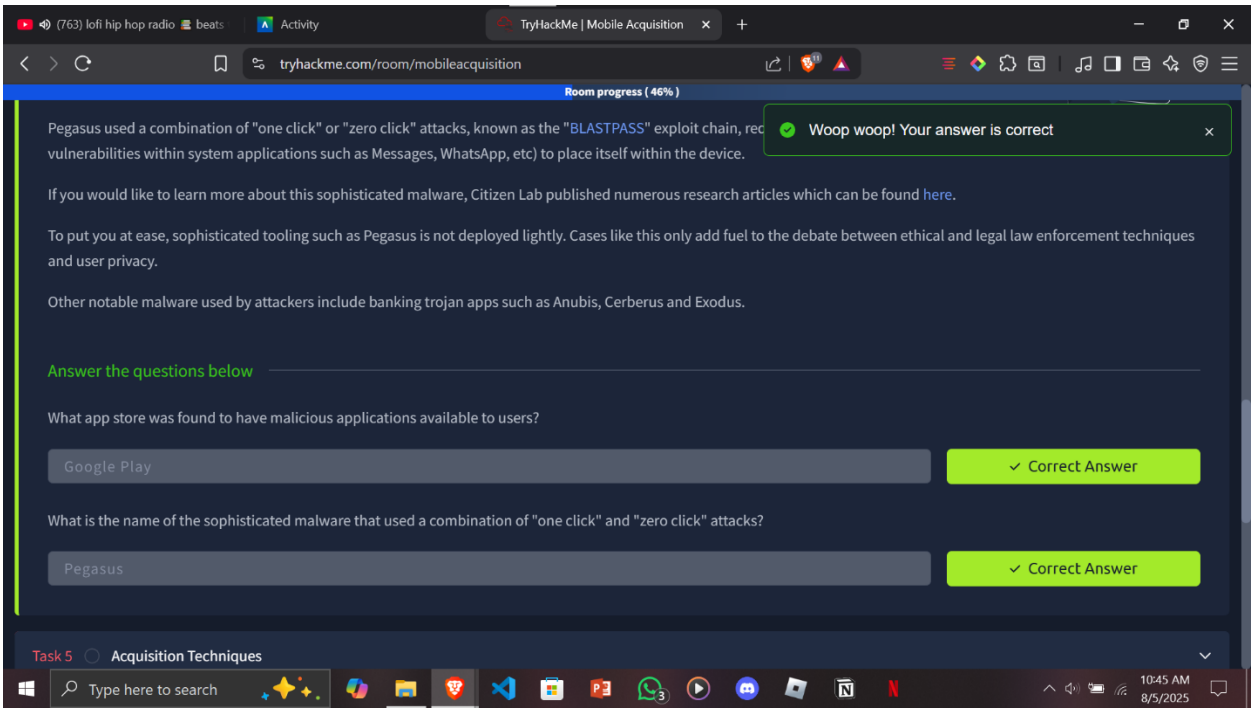
Key Takeaway

This exercise examines manufacturer-level protections like secure boot, which prevents untrusted code loading and hence limits forensic access. It also makes evident that encryption keys within modern phones are stored within hardware, making them difficult to extract and further reinforcing the need to employ vigilance when employing technique. Such limitations highlight the complexity of mobile data extraction within forensic procedures.

Task 4: APTs Meet Mobile Devices

In this task, you discover how mobile phones are vulnerable to cyberattacks, specifically from Advanced Persistent Threats (APTs). This features malware like Pegasus and discusses how vulnerabilities can be used by attackers with no user interaction. The task reflects the evolving face of mobile threats in both personal and political contexts.

Upon task completion, all stipulated questions were resolved with precision. This is evidenced in the subsequent figure:



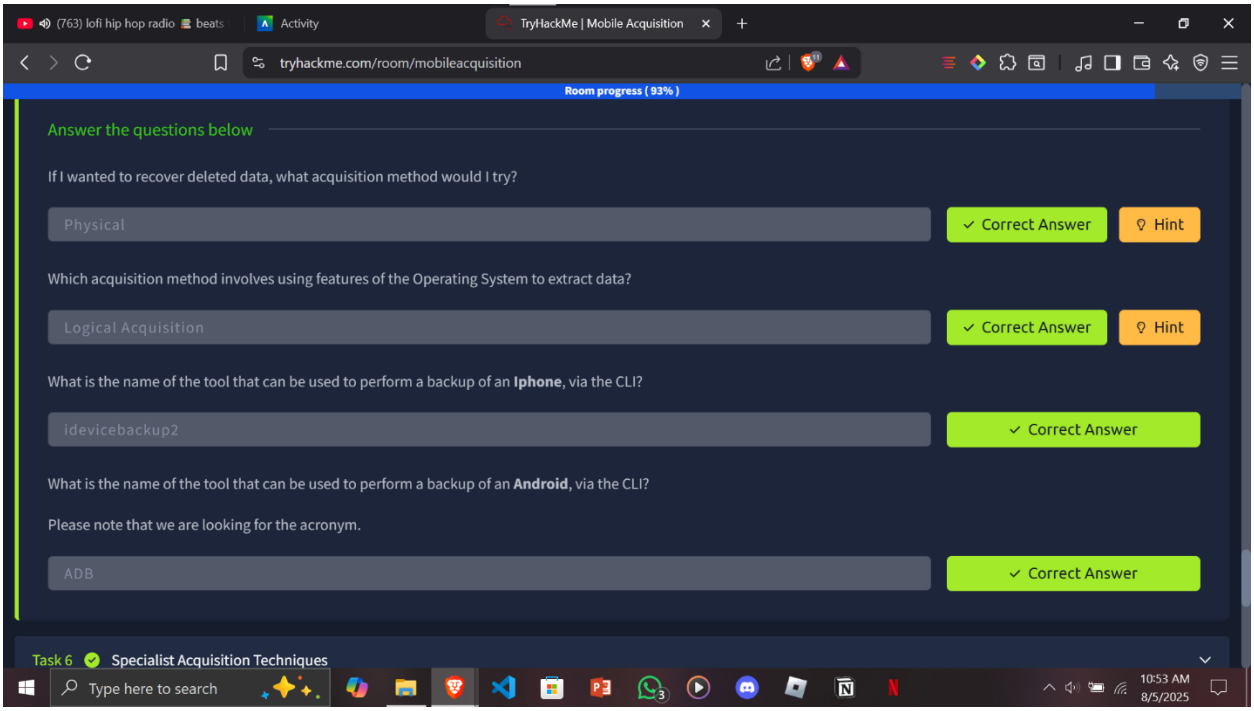
Key Takeaway

In this task, one learns that the Google Play Store has been found to harbor malicious applications, which goes to show that even legitimate marketplaces can become penetrated. The notorious Pegasus malware is featured, which uses one-click as well as zero-click exploits in its attempt to penetrate into mobile devices covertly. This emphasizes the growing sophistication of APT (Advanced Persistent Threat) in targeting mobile ecosystems.

Task 5: Acquisition Techniques

This assignment introduces the main methods of data acquisition from mobile devices: physical and logical acquisition. The assignment covers the strengths and weaknesses of the two methods, particularly with regard to data depth and forensic soundness. The assignment also introduces tools like ADB and idevicebackup2 for practical use on Android and iOS devices.

Following the successful execution of the task, the provided inquiries have been accurately addressed. This is then demonstrated in the accompanying illustration:



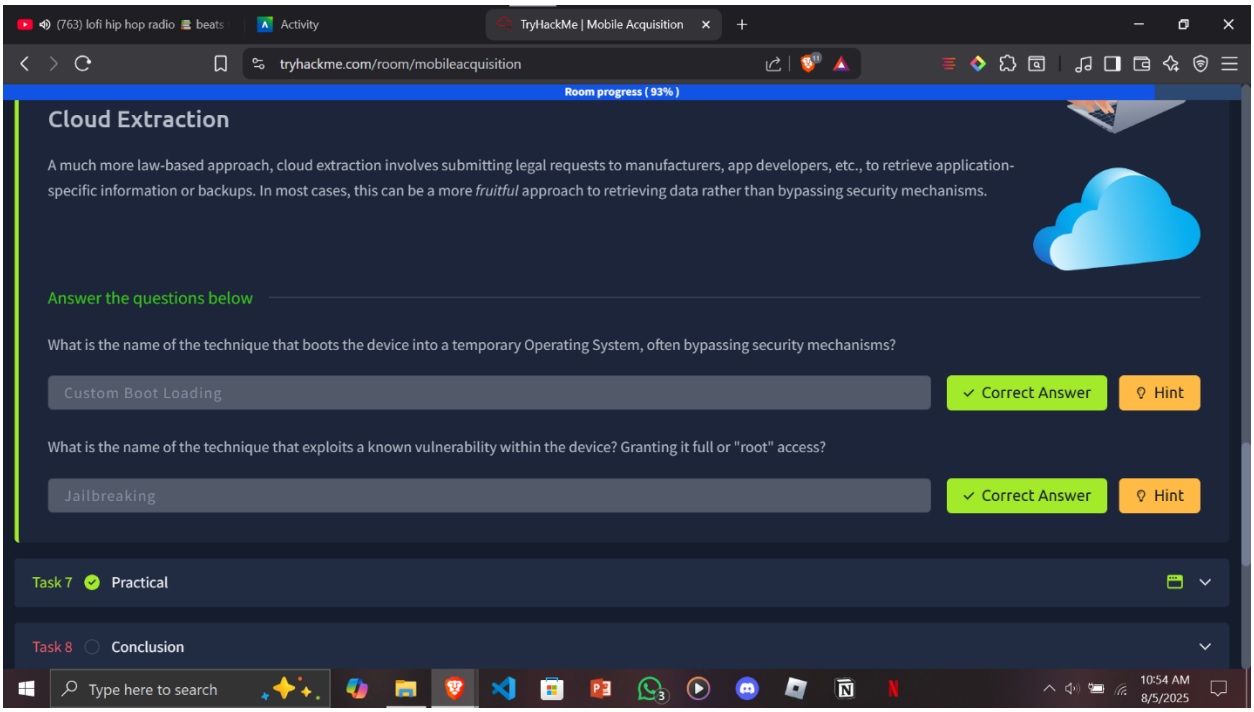
Key Takeaway

This task distinguishes between physical acquisition, which captures a bit-by-bit image (including erased data), and logical acquisition, which utilizes OS functionalities and preserves integrity without altering device information. It also touches on CLI tools like idevicebackup2 for iPhone backup and ADB for Android, mentioning their relevance in more secure extractions. Together, these methods represent compromises between recovery depth and forensic integrity.

Task 6: Specialist Acquisition Techniques

This task discusses advanced acquisition methods used when standard procedures are ineffective or impeded. It explains how advanced methods like jailbreaking and custom recovery modes enable greater levels of data access. These methods require greater skill and caution as they can potentially alter or destroy the evidence.

The successful fulfillment of the task yielded precise responses to the stipulated questions. This is empirically validated in the figure shown below:



Key Takeaway

You discuss advanced techniques: custom boot loading, loading a device to a temporary OS to circumvent maker limitations, and jailbreaking, using known bugs to gain root access. These are more intrusive techniques of gaining access to device internals but present higher forensic risk and technical skill. They illustrate the evolution from novice to expert-level forensic techniques.

Task 7: Practical

This final task presents a hands-on practical of the room Mobile Acquisition. It involves one taking the place of a forensic analyst on TryAnalyseMe's mobile analysis team. There is a new case and one's job is to:

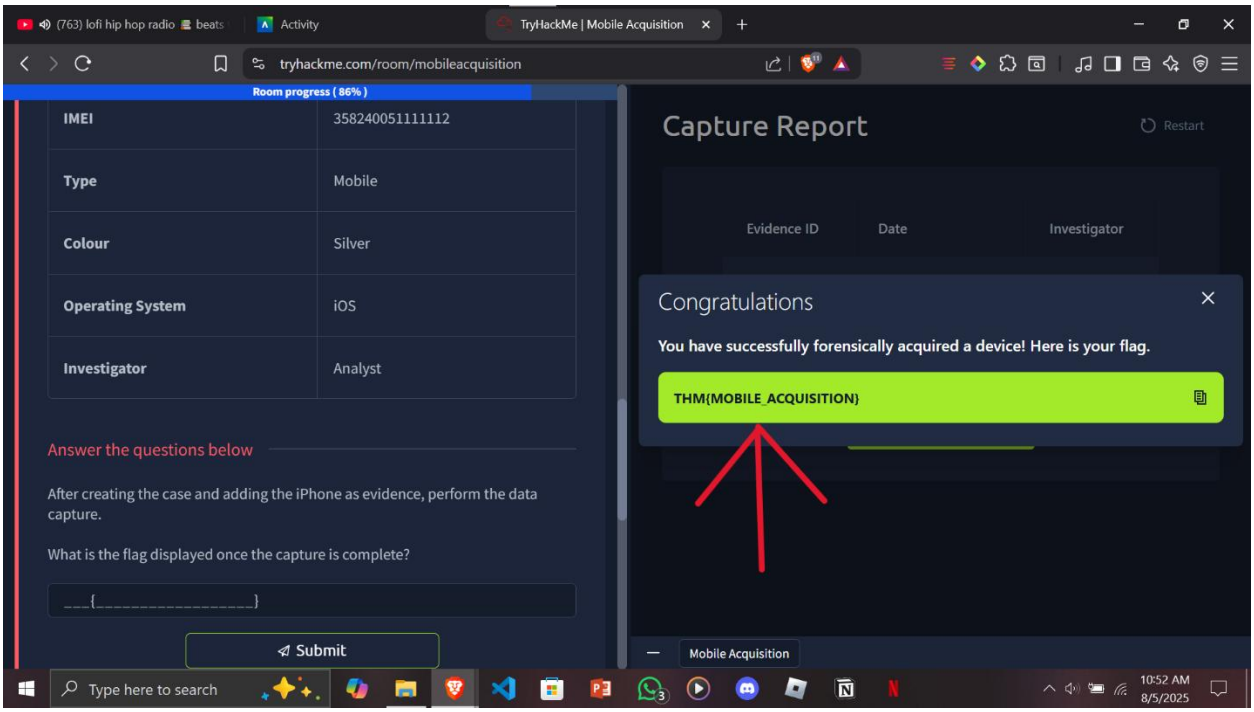
Create the case (As Investigator "Analyst");

add the evidential item using the details in the following table;

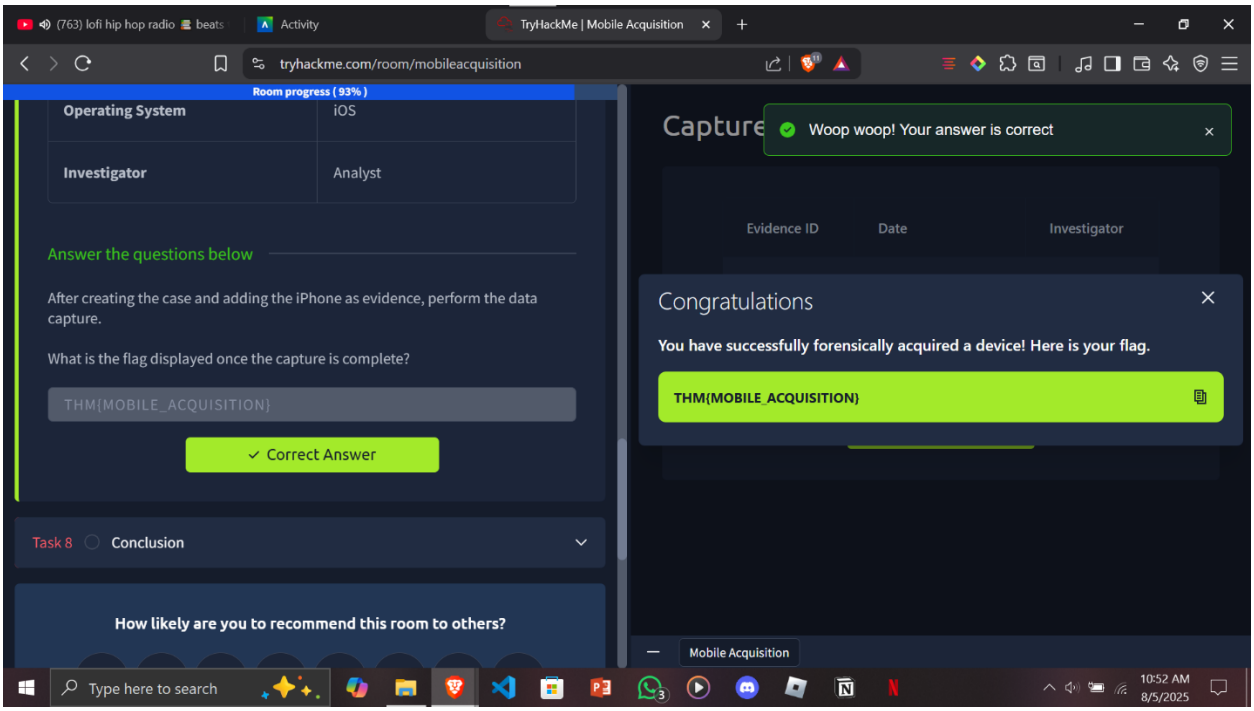
perform data acquisition and

export the report to retrieve the flag.

After performing the job, the flag was retrieved. This is shown in the image below:



The flag was then used to answer the question as shown in the image below. This as the main question of the practical.

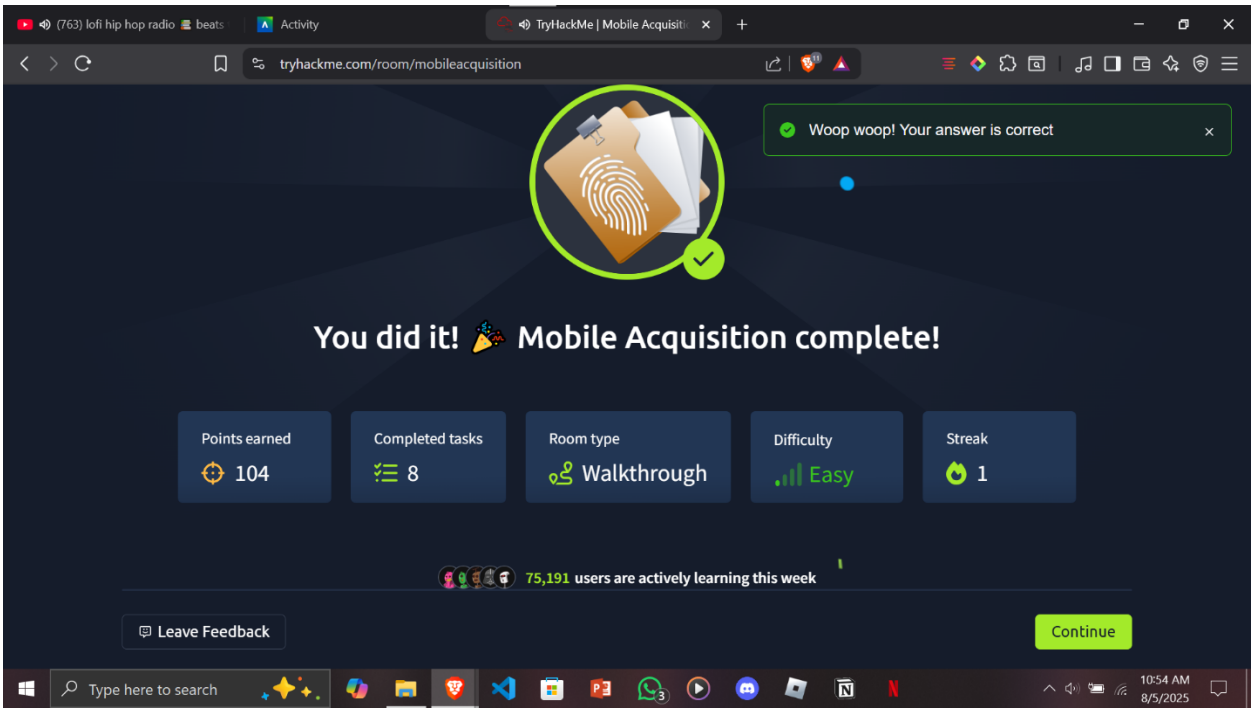


Key Takeaway

This is the hands-on component whereby you create a forensic case, add an iPhone as evidence, and perform an actual data capture using tools and techniques covered within the room. Compelling the process to success awards the flag "THM{MOBILE_ACQUISITION}", which denotes successful data extraction alongside accuracy in procedures. It combines theory with practice in a simulation of mobile forensic examination in real life.

Task 8: Conclusion

The Mobile Acquisition room is a top-level overview of how forensic examiners acquire and process data from cellphones. It begins by outlining the significance of smartphones as repositories of rich behavioral, personal, and communications data in today's investigations. The lecture analyzes the increasing interest of threat actors—especially Advanced Persistent Threats (APTs)—in mobile platforms and outlines the technical and legal barriers to access offered by device encryption, secure boot, and manufacturer protection. Students learn both fundamental and advanced methods of acquisition, such as logical and physical imaging, as well as advanced techniques like jailbreaking and custom bootloading. Finally, the room concludes with an experiential exercise in which participants are guided through carrying out a real mobile data capture, reinforcing theory as well as technical skills necessary for mobile device forensic analysis.



References

TryHackMe. (n.d.). *Mobile Acquisition* [Online training room]. TryHackMe. https://tryhackme.com/room/mobileacquisition